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(54) An improved stent which is easly recaptured and repositioned within the body

(57) In accordance with the present invention there is provided a pre-cursor stent for positioning within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries of a patient to assist in repairing the abdominal aortic aneurysm. The stent is designed to be coupled to a graft for directing blood flow through the aneurysm. The stent is made from a substantially cylindrical self-expanding member having a proximal end, a distal end, a longitudinal axis extending therebetween and an interior. The precursor stent further includes at least two spaced apart longitudinal legs having distal and proximal ends, the proximal ends of the legs attached to the distal end of the member, the legs extending distally from the member, each the leg including a flange adjacent its distal end. Also in accordance with the present invention is an aortic graft, having the above mentioned precursor stent attached thereto when fully deployed. Further in accordance with the present invention is a delivery apparatus for the above described precursor stent.

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Description

FIELD OF THE INVENTION

[0001] The invention relates to a precursor stent, and a delivery apparatus therefor, for use with an aortic graft for repairing an abdominal aortic aneurysm.

BACKGROUND OF THE INVENTION

[0002] An abdominal aortic aneurysm is a sac caused by an abnormal dilation of the wall of the aorta, a major artery of the body, as it passes through the abdomen. The abdomen is that portion of the body which lies between the thorax and the pelvis. It contains a cavity, known as the abdominal cavity, separated by the diaphragm from the thoracic cavity and lined with a membrane, the peritoneum. The aorta is the main trunk, or artery, from which the systemic arterial system proceeds. It arises from the left ventricle of the heart, passes upward, bends over and passes down through the thorax and through the abdomen to about the level of the fourth lumbar vertebra, where it divides into the two common iliac arteries.

[0003] The aneurysm usually arises in the infrarenal portion of the diseased aorta, for example, below the kidneys. When left untreated, the aneurysm will eventually cause rupture of the sac with ensuing fatal hemorrhaging in a very short time. High mortality associated with the rupture has led to the present state of the art and the trans-abdominal surgical repair of abdominal aortic aneurysms. Surgery involving the abdominal wall, however, is a major undertaking with associated high risks. There is considerable mortality and morbidity associated with this magnitude of surgical intervention, which in essence involves replacing the diseased and aneurysm segment of blood vessel with a prosthetic device which typically is a synthetic tube, or graft, usually fabricated of either DACRON®, TEFLON®, or other suitable material.

[0004] To perform the surgical procedure requires exposure of the aorta through an abdominal incision, which can extend from the rib cage to the pubis. The aorta must be closed both above and below the aneurysm, so that the aneurysm can then be opened and the thrombus, or blood clot, and arterioscleriotic debris removed. Small arterial branches from the back wall of the aorta are tied off The DACRON® tube, or graft, of approximately the same size of the normal aorta is sutured in place, thereby replacing the aneurysm. Blood flow is then reestablished through the graft. It is necessary to move the intestines in order to get to the back wall of the abdomen prior to clamping off the aorta.

[0005] If the surgery is performed prior to rupturing of the abdominal aorta aneurysm, the survival rate of treated patients is markedly higher than if the surgery is performed after the aneurysm ruptures, although the mortality rate is still quite high. If the surgery is performed prior to the aneurysm rupturing, the mortality rate is typically less than 5%. Conventional surgery performed after the rupture of the aneurysm is significantly higher, one study reporting a mortality rate of 66.7%. Although abdominal aortic aneurysms can be detected from routine examinations, the patient does not experience any pain from the condition. Thus, if the patient is not receiving routine examinations, it is possible that the aneurysm will progress to the rupture stage, wherein the mortality rates are significantly higher.

[0006] Disadvantages associated with the conventional, prior art surgery, in addition to the high mortality rate, are: the extended recovery period associated with such surgery, difficulties in suturing the graft, or tube, to the aorta, the loss of the existing thrombosis to support and reinforce the gruff; the unsuitability of the surgery for many patients having abdominal aortic aneurysms; and the problems associated with performing the surgery on an emergency basis after the and has ruptured. As to the extent of recovery, a patient can expect to spend from 1 to 2 weeks in the hospital after the surgery, a major portion of which is spent in the intensive care unit, and a convalescence period at home from 2 to 3 months, particularly if the patient has other illness such as heart, hmg, liver, and/or kidney disease, in which case the hospital stay is also lengthened. Since the graft must be secured, or sutured, to the remaining portion of the aorta, it is often difficult to perform the suturing step because of thrombosis present on the remaining portion of the aorta, and that remaining portion of the aorta wall may be friable, or easily crumbled.

[0007] Since the thrombosis is totally removed in the prior art surgery, the new graft does not have the benefit of the previously existing thrombosis therein, which could be utilized to support and reinforce the graft, were the graft to be able to be inserted within the existing thrombosis. Since many patients having abdominal aortic aneurysms have other chronic illnesses, such as heart, lung, liver, and/or kidney disease, coupled with the fact that many of these patients are older, the average age being approximately 67 years old, these patients are not ideal candidates for such surgery, which is considered major surgery. Such patients have difficulties in surviving the operation. Lastly, once the aneurysm has ruptured, it is difficult to perform a conventional surgery on an expedited basis because of the extent of the surgery.

[0008] Accordingly, the prior art teaches various methods and apparatus for repairing an abdominal aortic aneurysm which is believed to lower morbidity and mortality rate by not requiring an abdominal incision and general anesthesia, not requiring suturing the graft to the remaining aortic wall, and which permits the existing aortic wall and thrombosis therein to be retained to reinforce and support the aortic graft. An example of such a method and apparatus is given in U.S. Patents 5,316,023 issued to Palmaz et al. on May 31, 1994; 5,360,443 issued to Barone et al. on November 1, 1994;

5,578,071 issued to Parodi on November 26, 1996; and 5,591,229 issued to Parodi on January 7, 1997, all of which are hereby incorporated herein by reference.

[0009] Devices, such as the one shown in the above referenced Barone patent, use an improved method for repairing an abdominal aortic aneurysm in an aorta having two iliac arteries associated therewith. The device includes first and second tubes, preferably made from a variety of materials such as DACRON® and other polyester materials, TEFLON® (polytetrafluoroethylene), TEFLON® coated DACRON®, porous polyurethane, silicone, expanded polytetrafluoroethylene, and expanded polyurethane. It is preferred that all of the foregoing materials be porous to allow for an intimal layer to form on the tubes 160. Each of the tubes are connected to expandable and deformable, tubular members, or stents. These stents can be similar in structure to those described in disclosed in U.S. Patents 4,733,665 issued on March 29, 1988; U.S. Patent 4,739,762, issued on April 26, 1988; and U.S. Patent 4,776,337 issued on October 11, 1988, all of the foregoing patents being in the name of Julio C. Palmaz, each of which is incorporated herein by reference. Each of the tube/stent structures are then disposed on the end of a balloon catheter. Either both tubes are inserted into the same femoral artery or one of the tubes is inserted into one femoral artery of the patient and the other tube is inserted into the other femoral artery of the patient. Thereafter the tubes are intraluminally delivered to the aorta, thereby disposing at least a portion of each tube within the abdominal aortic aneurysm. The balloon catheters are then expanded to expand and deform the tubular members, to force the tubular members radially outwardly into contact with the aorta and each other. This secures the tubular members and a least a portion of each tube within the aorta, whereby the tubes provide a bilateral fluid passageway through the abdominal aortic aneurysm.

[0010] While the above mentioned devices would seem to work well, there is a desire to improve upon the device. More particularly, there was a need to ensure that most of the blood flowing through the abdomen, flows through the bilateral fluid passageways and not around them where it could cause further damage. The precursor stent gasket described in the above referenced parent case to this application, limits the amount of blood which could leak around the bilateral fluid passageways and into the aneurysm. The pre-cursor stent is positioned within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries, of a patient to assist in repairing the abdominal aortic aneurysm. The stent is designed to be coupled to the bilateral grafts for directing blood flow. The graft has a distal end for positioning distal to the aneurysm, and a proximal end for positioning proximal to the aneurysm. The precursor stent includes a substantially cylindrical expandable member having a proximal end, a distal end and an interior. The stent further includes a compressible gasket member located within the interior of the expandable member and attached thereto. The compressible member is substantially impervious to blood when in a compressed state. In addition, the stent has a means, within the compressible member, for coupling the graft to the gasket member. This is so the coupled device can direct blood flow through the graft, with the gasket member substantially preventing blood from flowing into the aneurysm.

[0011] While the above described precursor stent gasket works well, there has been a desire to improve upon it. There has been a desire to design an improved gasket which can be more accurately placed within the body by making the gasket retrievable back into the delivery device if the physician determines that his current placement is not ideal. That is, a retrievable and re-positional stent would be advantageous. While retrievable stents are known in the art, none of them allow the body or member of the stem to be fully deployed and then recapture without having the stent attached to a separate line or the like. In addition, in the past making the stent gasket retrievable posed design difficulties. The retrievable features should not interfere with the gasket material. Puncturing the gasket material might lead to blood leakage within the aneurysm. In addition, the small size of the precursor stent gasket and its delivery system poses some difficulties with retrievability. The present invention not only overcomes these problems associated with prior precursor stents, but provides a advantageous repositionable and re-capturable stent that can have many applications other than abdominal aortic aneurysms.

SUMMARY OF THE INVENTION

[0012] In accordance with the present invention there is provided a pre-cursor stent for positioning within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries of a patient to assist in repairing the abdominal aortic aneurysm. The stent is designed to be coupled to a graft for directing blood flow through the aneurysm. The stent is made from a substantially cylindrical self-expanding member having a proximal end, a distal end, a longitudinal axis extending therebetween and an interior. The precursor stent further includes at least two spaced apart longitudinal legs having distal and proximal ends, the distal ends of the legs are attached to the proximal end of the member, the legs extending proximally away from the member, each the leg including a flange adjacent its proximal end.

[0013] Further in accordance with the present invention there is provided an aortic graft for intravascular delivery to repair an abdominal aortic aneurysm in an aorta having two iliac arteries associated therewith. The graft includes first and second graft members each having distal and proximal ends and each designed to be inserted through a femoral artery in a collapsed condition, and inserted within the aneurysm and deployed therein. The distal ends of the graft members are distal to the

aneurysm adjacent an arterial wall. The aortic graft further includes a precursor stent, surrounding the distal ends of each of graft member. The precursor stent is made from a substantially cylindrical self-expanding member having a proximal end, a distal end, a longitudinal axis extending therebetween and an interior. The precursor stent includes at least two spaced apart longitudinal legs having distal and proximal ends. The distal ends of the legs are attached to the proximal end of the member, and the legs extending proximally away from the member. Each of the legs includes a flange adjacent its distal end. The precursor stent further includes a gasket member attached thereto for substantially preventing blood from flowing through any gaps between the distal ends of the graft members, and between the distal ends of the graft members and the arterial wall.

[0014] Even further in accordance with the present invention, there is provided a delivery apparatus for a self-expanding stent. The apparatus includes an outer sheath, comprising an elongated tubular member having distal and proximal ends, and an inner shaft located coaxially within the outer sheath, the shaft having a distal end and a proximal end. The distal end of the shaft further including at least two grooves disposed thereon. The apparatus further includes a substantially cylindrical self-expanding stent located within the sheath and making frictional contact therewith. The self-expanding member has a proximal end, a distal end, a longitudinal axis extending therebetween and an interior. The self-expanding stent further includes at least two spaced apart longitudinal legs having distal and proximal ends. The distal ends of the legs are attached to the proximal end of the member, and the legs extend proximally away from the member. Each of the legs includes a flange adjacent its proximal end. The flanges are set within the grooves of the inner shaft, thereby releasable attaching the stent to the inner shaft.

BRIEF DESCRIPTION OF DRAWINGS

[0015] The foregoing and other aspects of the present invention will best be appreciated with reference to the detailed description of the invention in conjunction with the accompanying drawings, wherein:

[0016] Figure 1 is a perspective view of one preferred embodiment of precursor stent 10 made in accordance with the present invention having the gasket member 30 removed for clarity.

[0017] Figure 2 is a partial exploded view of the distal end of apparatus 40, made in accordance with the present invention, and the legs and flanges of the precursor stent.

[0018] Figure 3 is a partial cross sectional view of an abdominal aortic aneurysm showing apparatus 40, made in accordance with the present invention, inserted therein with precursor stent in its fully undeployed position.

[0019] Figure 4 is a view similar to that of Figure 3 but showing the precursor stent in its partially deployed position.

[0020] Figure 5 is a view similar to that of Figure 4 but showing the precursor stent in its fully deployed position.

[0021] Figure 6 is a partial cross sectional view taken along line 6-6 of Figure 5.

[0022] Figure 7 is a view similar to that of Figure 5 but showing the precursor stent 10 coupled to endografts.

[0023] Figure 8 is a is a partial cross-sectional view a prior art bilateral intra-aortic bypass graft of the type to be used with the present invention.

DETAILED DESCRIPTION OF THE INVENTION

[0024] The present invention is designed to be coupled and/or used with a graft for directing blood flow. Referring now to the drawings, wherein like numerals indicate the same element throughout the views, there is shown in Figure 8, a prior art version of such a graft. The type of graft it is designed to be coupled to is very similar to types of grafts known in the prior art. Therefore, a description of a prior art graft may be helpful. Figure 8 shows such a graft. Figure 8 shows a bilateral intra-aortic bypass graft 150 for intraluminal delivery to repair an abdominal aortic aneurysm 151 in an aorta 152 having two iliac arteries 153L, 153R associated therewith. Associated with aorta 152, above aneurysm 151, are a plurality of renal arteries 157, in fluid communication with aorta 152. Bilateral intra-aortic bypass graft 150, as well as other grafts to be hereinafter described, could also be utilized in the thoracic aorta, and can be used to repair thoracic aneurysms or thoracic dissecting aneurysms. Accordingly, use of the term "aortic aneurysm" in this specification and claims is intended to relate to and mean both abdominal aortic aneurysms and thoracic aneurysms.

[0025] Bypass graft 150 is seen to generally comprise a first graft tube 160A having distal and proximal ends 172A and 173A, at least a portion of the graft tube 160A adapted to be disposed within the aneurysm 151, preferably so that its distal end is distal to the aneurysm and its proximal end is proximal to the aneurysm. A second graft tube 160B is similarly situated on the right side. Graft 150 also includes first and second tubular stent members 162A, 162B, each having proximal and distal ends 181A & 181B, and 182A & 182B located within grafts 160. Each stent member 162A, 162B has proximal and distal ends, preferably positioned so that the distal ends are distal to the aneurysm and the proximal ends are proximal to the aneurysm.

[0026] The stent members 162, along with graft tubes 160 permit intraluminal delivery into the aorta 152. This is accomplished by percutaneously inserting the stent members into the same or different femoral arteries and navigating them into the aorta. This type of procedure is similar to delivery of angioplasty catheters and guiding catheters into the human vasculature. Upon the place-

ment of the stem members they are deployed either through a radially, outwardly extending force, such as a balloon catheter, or self-expanding stents and deployed by releasing the stent members from a constraint. Once deployed, a bilateral passageway is formed within the abdominal aortic aneurysm by passageways 191A, 191B extending through the stent members 162 and graft tubes 160 forming a generally inverted Y-shaped configuration. Each stent member 162A, 162B preferably has a smooth outer wall surface disposed between its distal and proximal ends. Stent members 162 preferably have a substantially uniform thickness with a plurality of slots formed therein.

[0027] Graft tubes 160A, 160B preferably have a generally, circular cross-sectional configuration, and can be made from a variety of materials, provided they have the requisite strength characteristics to be utilized as a bypass graft 150, as well as have the requisite compatibility with the human body in order to be used as a graft, or implant material, without being rejected by the patient's body. Examples for such materials are DACRON Registered TM and other polyester materials, TEFLON Registered TM (polytetrafluoroethylene), TEFLON Registered TM coated DACRON Registered TM , porous polyurethane, silicone, expanded polytetrafluoroethylene, and expanded polyurethane. It is preferred that all of the foregoing materials be porous to allow for an intimal lays to form on the graft tubes 160. Additionally, graft tubes 160A, 160B can be made by the replamineform replicated life forms process, which is a method for fabricating uniformly microporous materials from marine skeletal structures. The foregoing described fabric materials can be knitted or woven, and can be warp or weft knitted. If the material is warp knitted, it may be provided with a velour, or towel like surface, which speeds up clotting of blood which contacts graft tubes 160A, 160B in order to increase the attachment, or integration, of graft tubes 160A, 160B to aorta 152, or to assist the integration of graft tubes 160A, 160B to the thrombosis 154. Graft tubes 160A, 160B can also be made of a biodegradable, or degradable material, such as albumin or collagen or a collagen coated material. A graft tube which is bioerodible, would erode and dissolve, or degrade, over a period of time; however, it is believed that a layer of endothelium, or skin, will grow as the graft tubes 160A, 160B erode, the new layers of endothelium, or skin, provide a new, fluid impervious lining within aneurysm 151. In some procedures, it might be desirable to make graft tubes 160A, 160B of a fluid impervious material. Additionally, graft tubes 160A, 160B or stent 162A, 162B, could have a coating of a biologically inert material, such as TEFLON Registered TM or porous polyurethane.

[0028] If any of the foregoing described materials are used for the manufacture of graft tubes 160, the graft tubes may be connected to the stent members 162 as by a plurality of conventional sutures of polypropylene, DACRON Registered TM , or any other suitable material. Preferably, the ends of graft tubes 160 overlap and fully cover the second ends of stent members 162, such overlapping being approximately 100% of the length of stent members 162.

[0029] The present invention improves upon the prior art graft 150, mentioned above, by further including, and preferably initially deploying, a precursor stent 10, shown in Figure 1. Stent 10 is to be deployed within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries of a patient to assist in repairing the abdominal aortic aneurysm. The stent is designed to be coupled to a graft, such as the one described above, for directing blood flow through the aneurysm. The stent is made from a substantially cylindrical self-expanding member 12 having a proximal end 14, a distal end 16, a longitudinal axis extending therebetween and an interior 18. The precursor stent further includes at least two, but preferably 8 as shown in Figure 1, spaced apart longitudinal legs 20 each having proximal and distal ends 24 and 26 respectively. Preferably, there is a leg extending from each apex 11 of diamonds 13. The distal ends 26 of the legs are attached to the proximal end 14 of the member 12, the legs extending proximally away from the member. At least one, but preferably each leg includes a flange 28 adjacent its proximal end which, as is described in greater detail below, allows for the stent to be retrievable into its delivery apparatus after partial or full deployment of member 12 so that it can be turned, or otherwise repositioned for proper alignment.

[0030] Self expanding stents are typically made from superelastic Nickel Titanium alloys (Nitinol). Descriptions of medical devices which use such alloys can be found in U.S. Patents 4,665,906 issued to Jervis on May 19, 1987, which is hereby incorporated herein by reference. Stent 10 is preferably laser cut from a tubular piece of Nickel Titanium Alloy and thereafter treated so as to exhibit superelastic properties at body temperature. Stent 10 is shown in the figures as being a zigzag or diamond patterned stent, having approximately 8 diamonds, and when the stent is fully expanded the diamonds would have angles of 45-55 degrees at their distal and proximal ends. However, stent 10 can take on many different patterns or configurations.

[0031] Many of the advantages of stent 10 can be better understood by referring to its delivery apparatus 40 shown in Figures 2 and 3. Apparatus 40 includes an outer sheath 50 which is essentially an elongated tubular member, similar to ordinary guiding catheters which are well known to those of ordinary skill in the art. Sheath 50 has a distal end 52 and a proximal end (not shown) Apparatus 40 also includes an inner shaft 60 located coaxially within the outer sheath 50 prior to deployment as shown in Figure 3. The inner shaft has a distal end 52 and a proximal end (not shown). The distal end 52 of the shaft has at least two, preferably 8 to match the number of longitudinal arms and diamond apexes on stent 10, grooves 54 disposed thereon. As seen from Figure 3, when the apparatus is not fully deployed, stent

10 located within the sheath and making frictional contact therewith. The flanges on the legs of the stent are set within the grooves of the inner shaft, thereby releasable attaching the stent to the inner shaft.

[0032] The advantages of the longitudinal legs, its flanges and the grooves on the inner shaft can best be described by referring to Figures 3 and 4. Figure 3 shows the apparatus 40 with the stent in its fully un-deployed position. Figure 3 also shows an aorta 2, an abdominal aortic aneurysm 3, renal arteries 4A and 4B, and iliac vessels 6A and 6B of a human patient. As seen from Figure 3, the apparatus 40 has been percutaneously inserted into the femoral artery and guided within the patients vascular system and inserted into the aneurysm 3. As mentioned above, expandable member 12 of stent 10 is designed to be deployed within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries of a patient to assist in repairing the abdominal aortic aneurysm. As will become apparent below when discussing the gasket aspect of the present invention, placement of expandable member 12 is important. The physician must have precise placement of the member to ensure adequate repair of the aneurysm.

[0033] As seen from Figure 4, the present invention allows the physician to fully deploy member 12 within the body without fully releasing the entire stent 10 from engagement with the delivery device. The legs 20 of the stent interlock with grooves 54 on inner shaft 60. Therefore, if the physician decides that the placement of the stent as shown in Figure 4 is incorrect, he would then push on the outer member of the apparatus while keeping the inner member stationary, thereby resulting in the stent being retrieved or retracted within outer sheath 50 so that the physician could reposition the stent. The legs allow the physician to see how the member 12 would be position when fully deployed. Once the physician has good position of the member 12, the legs 20 are released form their engagement with the inner shaft and move away from member 12 so as not to interfere with the rest of the procedure, as shown by the arrows in Figure 5. The legs are very pliable and d not need to be pushable, so that they are as atraumatic as possible.

[0034] In order to prevent the physician from prematurely completely deploying the stent 10, a releasable stop is preferably placed on the inner shaft. The stop could be a ring having a greater diameter than the outer member, so that as the outer member is pulled proximally along the inner shaft it hits the stop, and prevents full deployment of the entire stent 10. The stop is preferably releasable attached to the inner member, by threads or the like, so that it can be released from its engagement with the inner shaft to allow the outer member to slide back enough to fully deploy the entire stent 10 within the body.

[0035] As seen from the figures, the flanges 28 are substantially I-shaped. However, the flanges can be of any suitable shape. As used herein, flange means any protrusion on the proximal ends of legs 20, which is capable of interlocking with the grooves 54 on shaft 60. The I-shaped protrusions protrude from the legs in an axial direction. However, the flanges could point towards the interior of the expandable, much like a hook, and fit within a deep groove on the shaft. In addition, it is preferred that the legs of the stent be equally spaced about the distal end of the expandable member. This is so the stent is uniformly retracted within the sheath upon it being retrieved.

[0036] In one embodiment of precursor stent 10, shown in most of the figures but removed from Figure 1 for clarity, precursor stent 10 further includes a gasket member 30. This feature can be better understood by referring to Figure 6. As seen from that figure, precursor stent 10 further includes a gasket member 30. Gasket member 30 can be located within the expandable member such that it would come in the way of or impede any blood trying to flow through the interior of the expandable member or around the stent itself. For this embodiment gasket member 30 is a compressible member located within the interior 26 of the expandable member 12 and also covering the exterior of the stent as well. For the embodiment shown in Figure 6, gasket member 30 can be made from any number of materials known to those of ordinary skill in the art including open cell foam materials such as polyurethane, polyethylene, polytetrafluroethylene, other various polymer materials which are woven or knitted to provide a flexible structure such as Dacron, polyurethane, polypropylene, polytetrafluroethylene can also be used. Gasket 30 can be attached to expandable member 12 by any number of means including a plurality of conventional sutures of polypropylene, DACRON®, or any other suitable material and attached thereto. Other methods of attaching gasket 30 to expandable member include adhesives, ultrasonic welding, mechanical interference fit.

[0037] As will be explained later herein, it is preferable that the compressible member is substantially impervious to the flow of blood, at least when in a partially compressed state. When used throughout for the present invention, materials which are substantially impervious to the flow of blood include materials which become substantially impervious to the flow of blood after being saturated with blood. When the stent tubes and graft members, described above, are inserted and expanded within the gasket 30, the gasket 30 will compress. In this state, the gasket should be substantially impervious to blood so as to prevent blood from flowing through the interior 26 of member 12 and into the aneurysm.

[0038] The stent should include, within the compressible member, a coupling for joining a bilateral graft, such as graft 150, to the gasket member. As seen from Figure 3, gasket member 30 includes two substantially cylindrical conduits (although they could have any suitable shape such as a semi-cylindrical or D-shape cross-section), 32 and 34, extending through gasket 30. Conduits 32 and 34 are designed to receive one half of a bilateral graft in its un-expanded condition. After the grafts are

inserted into the conduits, they are expanded so that they are attached to stent 10. However, conduits 32 and 34 are not the only coupling for joining a bilateral graft, such as graft 150, to the gasket member. The coupling could be an integral part of the material the gasket 30 is made from. For example if gasket 30 is made from an open cell foam, the bilateral graft could pierce the material so as to effectively create its own conduit through the gasket 30. The coupling does not have to be a physical attachment, but rather some means for allowing the stents and grafts to work in operational engagement. This coupling is so that the combined precursor stent and bilateral graft direct blood flow through the graft, with the gasket member substantially preventing blood from flowing into the aneurysm.

[0039] Other alternative embodiments for the gasket member include attaching a compressible gasket member, similar to a drum gasket, to the distal end of the expandable member. Such drum gasket can be made from any number of materials known to those of ordinary skill in the art including various polymer materials which are woven, knitted, or foamed to provide a flexible structure such as polyurethane, polyethylene, polytetrafluroethylene, other various polymer materials which are woven or knitted to provide a flexible structure such as Dacron, polyurethane, polypropylene, polytetrafluroethylene can also be used. Such drum gasket can be attached to the expandable member by any number of means including a plurality of conventional sutures of polypropylene, DACRON®, or any other suitable material and attached thereto. A means for joining the bilateral graft to the drum gasket could include two substantially circular holes extending through the gasket or could also be an integral part of the material the gasket is made from. For example if the drum gasket is made from an open cell foam the bilateral graft could be pierce the material so as to effectively create its own conduit through the gasket. This coupling is so that the combined precursor stent and bilateral graft direct blood flow through the graft, with the gasket member substantially preventing blood from flowing into the aneurysm. In one particular embodiment a gasket of a drum type is placed within the stent at a predetermined distance from the distal end of the stent. The gasket would have two conduits, however, one would be larger than the other, with the smaller one being prdoaded onto a delivery system having a guidewire thereon. The preloaded guidewire would be used to guide one of the endografts into the stent. Thereafter, when a second guidewire is introduced for delivery of a second endograft, the guidewire will be blocked on one side by the drum gasket and guided into the empty and proper side.

[0040] After the precursor stent 10 has been deployed, bilateral grafts, similar to that shown in Figure 8, can then be deployed. Figure 7 shows how aneurysm 3 would appear after precursor stent has been fully deployed and two endografts 80 and 90 have been fully deployed as well. Since grafts 80 and 90 are substantially identical, a detailed description of a single endograft, graft 80, will now be given. Endograft 80 has a similar function but a different construction than the graft tube 160 and stent member 162 combination described above. Endograft 80 is preferably comprises a stent 81 laser cut from a tube of material, such as nitinol. The stent has a distal anchoring portion 82, a proximal anchoring stent 84, an middle portion 85 (not shown) and a flexible graft tube 86 covering the middle portion and attached thereto. The stent 81 is expandable from a compressed state to its implanted size. Distal anchoring portion 82 is designed to sealably contact and attach itself to the gasket member 30, and could include legs and flanges attached thereto, so as to make the stent 81 retractable and repositionable similar to stent 10. Proximal anchoring portion 84 is designed to be expanded so as to make contact with and attach itself to iliac artery 6A. Stents 81 is preferably a self-expanding stent, but could also be a plastically deformable balloon expandable stent, both types being discussed above. Graft tube 86 can be made from any of the materials graft member 160 can be made from. Preferred materials include a polymer material woven , spun, knitted, or other fabrication process obvious to those familiar with the art. Graft tube 86 is preferably impermeable to the flow of blood or becomes impermeable to blood flow therethrough after it is saturated. Graft tube 86 must be flexible to contour to the anatomy and of sufficient strength to sustain physiological blood pressure.

[0041] Figure 7 is a good illustration of how the present invention substantially prevents blood from flowing around endografts 80 and 90 and into the abdomen. As seen from that figure, expandable member 12 makes contact with the aorta 2 when it is expanded, and gasket member 30 fills the space between the bilateral endografts 80 and 90 and the aorta 2 this creating a seal which directs substantially all of the blood flow through the endografts.

[0042] Although particular embodiments of the present invention have been shown and described, modification may be made to the device and/or method without departing from the spirit and scope of the present invention. The terms used in describing the invention are used in their descriptive sense and not as terms of limitations.

Claims

1. A stent comprising:

(a) a substantially cylindrical self-expanding member having a proximal end, a distal end, a longitudinal axis extending therebetween and an interior; and

(b) at least two spaced apart longitudinal legs having distal and proximal ends, said distal ends of said legs being attached to said proxi-

mal end of said member, said legs extending proximally away from said member and at least one of said legs further including a flange adjacent its proximal end.

2. The stent of claim 1, wherein each of said legs includes a flange adjacent its proximal end.

3. The stent of claim 1 or claim 2, wherein said self-expanding member is made from a superelastic Nickel-Titanium alloy.

4. The stent of any one of claims 1 to 3, wherein said legs extend distally and axially from said member when said stent is deployed within a body.

5. The stent of any one of claims 1 to 4, wherein said longitudinal legs are equally spaced about said proximal end of said expandable member.

6. The stent of any one of claims 1 to 5, wherein said at least one flange on said longitudinal legs are substantially I-shaped.

7. The stent of any one of claims 1 to 6, which is retractable and repositionable and for implantation within the body of a patient.

8. The stent of any one of claims 1 to 6, which is adapted for positioning within the infrarenal neck, between an abdominal aortic aneurysm and the renal arteries of a patient, to assist in repairing the abdominal aortic aneurysm, said stent being designed to be coupled to a graft for directing blood flow through the aneurysm.

9. The stent of claim 8, wherein: self-expanding member further includes a gasket member which is substantially impervious to blood flow; and said gasket member is attached to said self-expanding member such that it will impede blood flow through said interior of said self-expanding member.

10. The stent of claim 9, wherein said gasket member comprises a compressible member located with said interior of said self-expanding member.

11. The stent of claim 9 or claim 10, wherein said gasket member comprises an open cell foam.

12. An aortic graft for intravascular delivery to repair an abdominal aortic aneurysm in an aorta having two iliac arteries associated therewith, said graft comprising:

(a) first and second graft members having distal and proximal ends, each graft member being designed to be inserted through a femoral artery in a collapsed condition, and inserted within the aneurysm and deployed therein so that said distal ends of said graft members are distal to said aneurysm adjacent an arterial wall; and
(b) a stent as defined in any one of claims 1 to 8, surrounding said distal ends of each of said graft members and coupled thereto, said stent further including a gasket member, optionally as defined in any one of claims 9 to 11, attached thereto for substantially preventing blood from flowing through any gaps between said distal ends of said graft members, and between said distal ends of said graft members and said arterial wall.

13. A delivery apparatus for a self-expanding stent, said apparatus comprising:

(a) an outer sheath, comprising an elongated tubular member having distal and proximal ends;
(b) an inner shaft located coaxially within said outer sheath, said shaft having a distal end and a proximal end, said distal end of said shaft further including at least two grooves disposed thereon; and
(c) a stent as defined in any one of claims 1 to 11 located within said sheath, wherein said flanges are set within said grooves of said inner shaft, thereby releasably attaching said stent to said inner shaft.

14. The apparatus of claim 13, wherein said flanges and said grooves are T-shaped.

15. The apparatus of claim 13 or claim 14, wherein said flanges fit completely within said grooves so that said stent adds no additional outside diameter size to said inner shaft.

FIG. 1

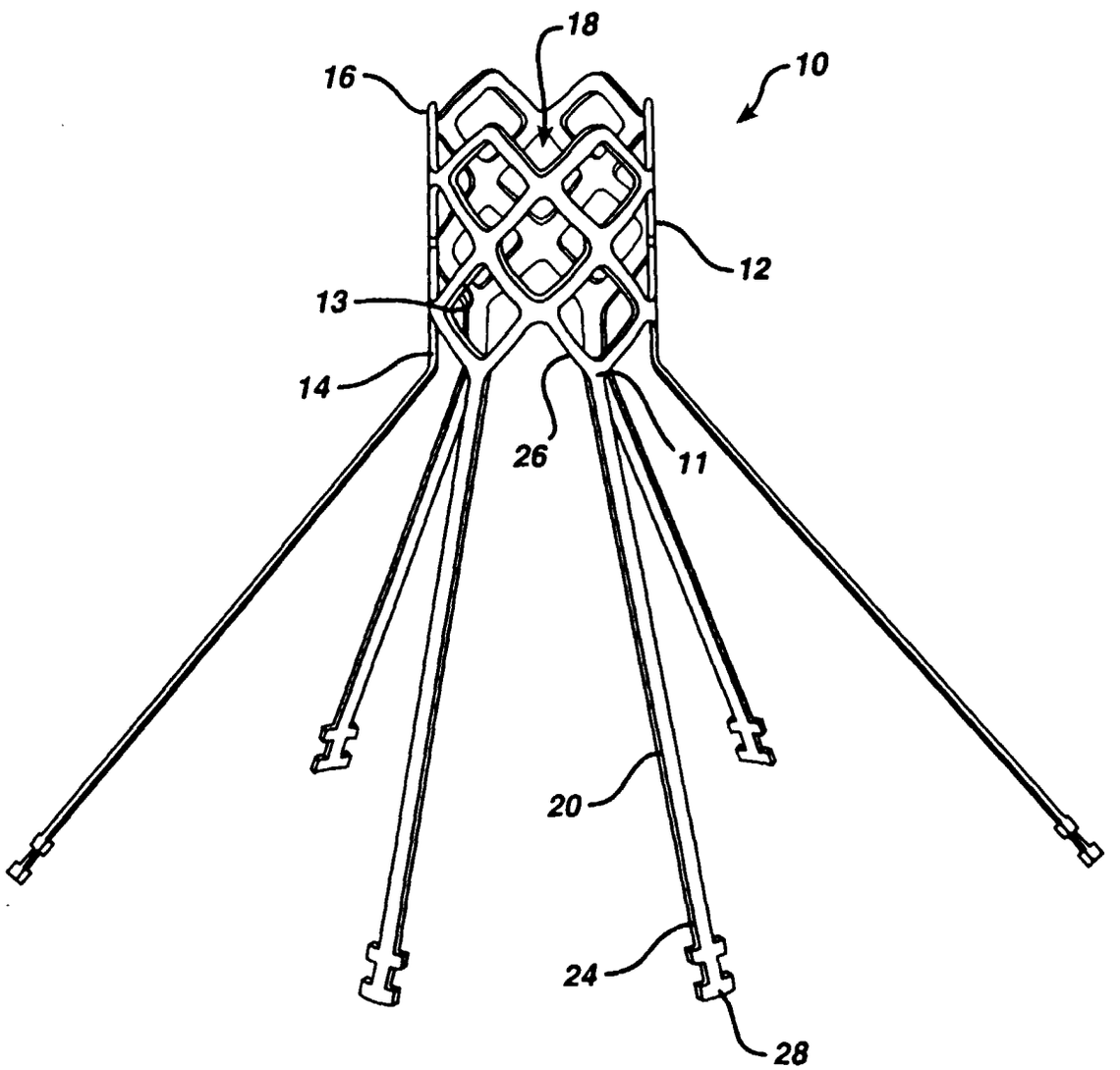


FIG. 2

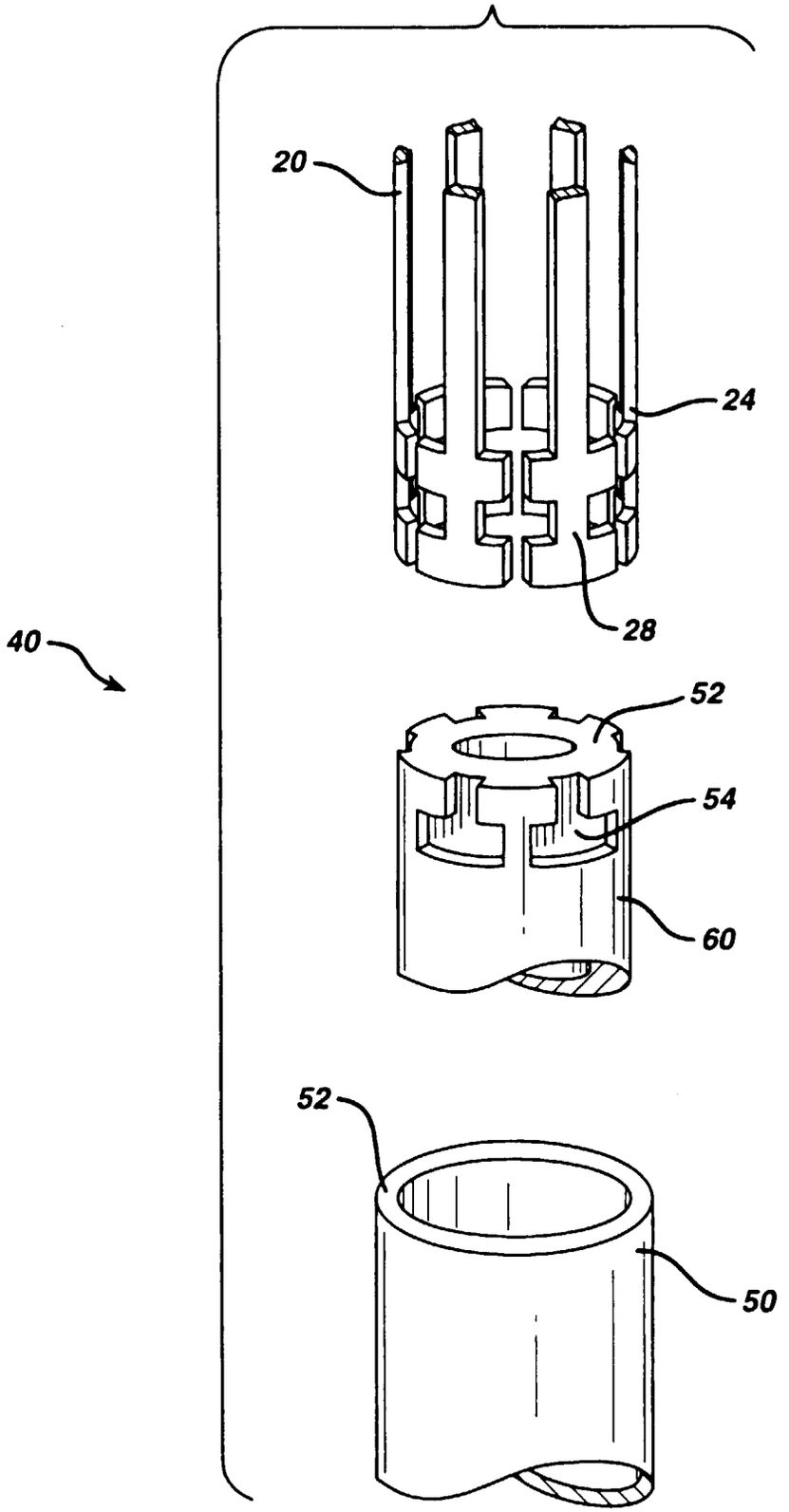


FIG. 3

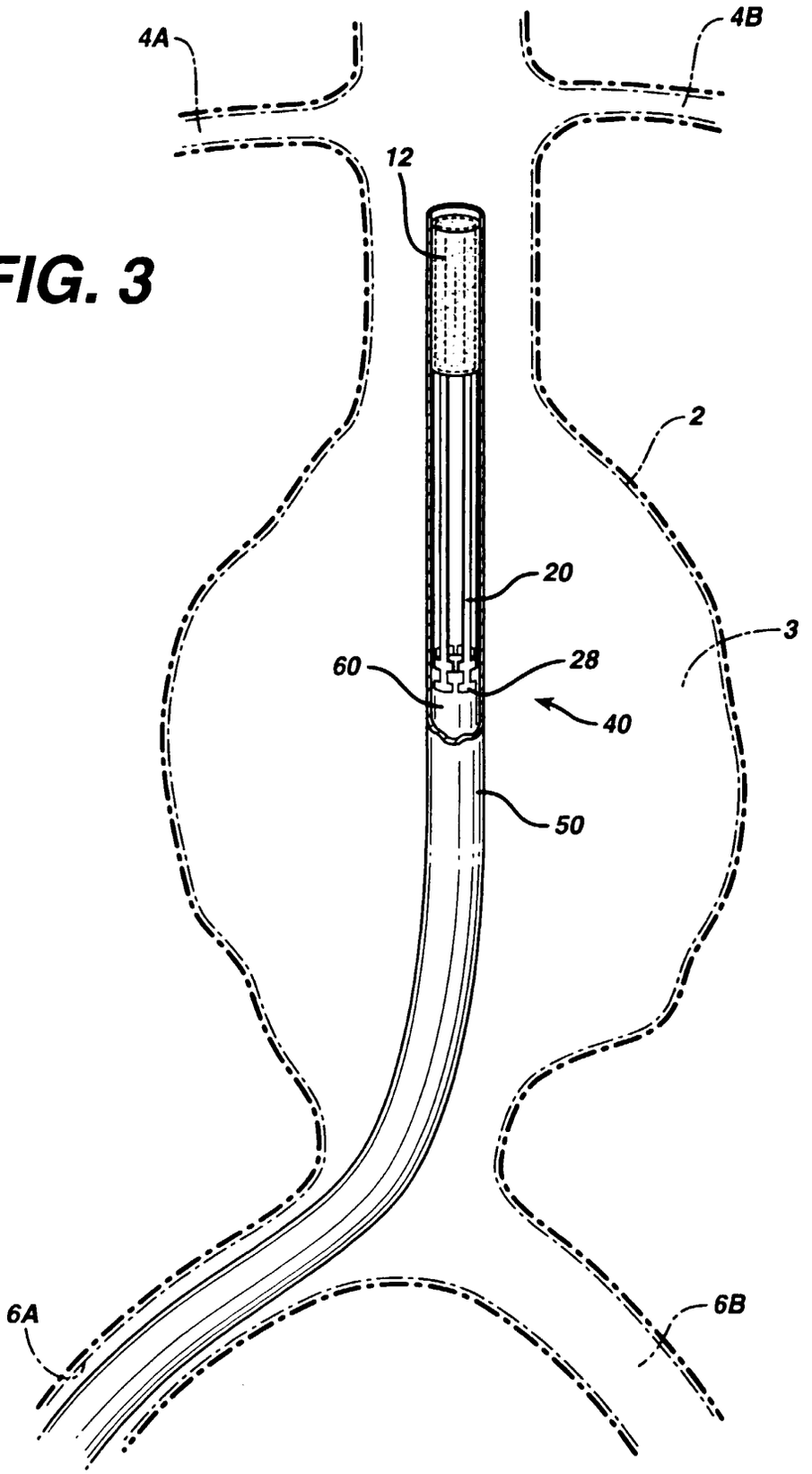


FIG. 4

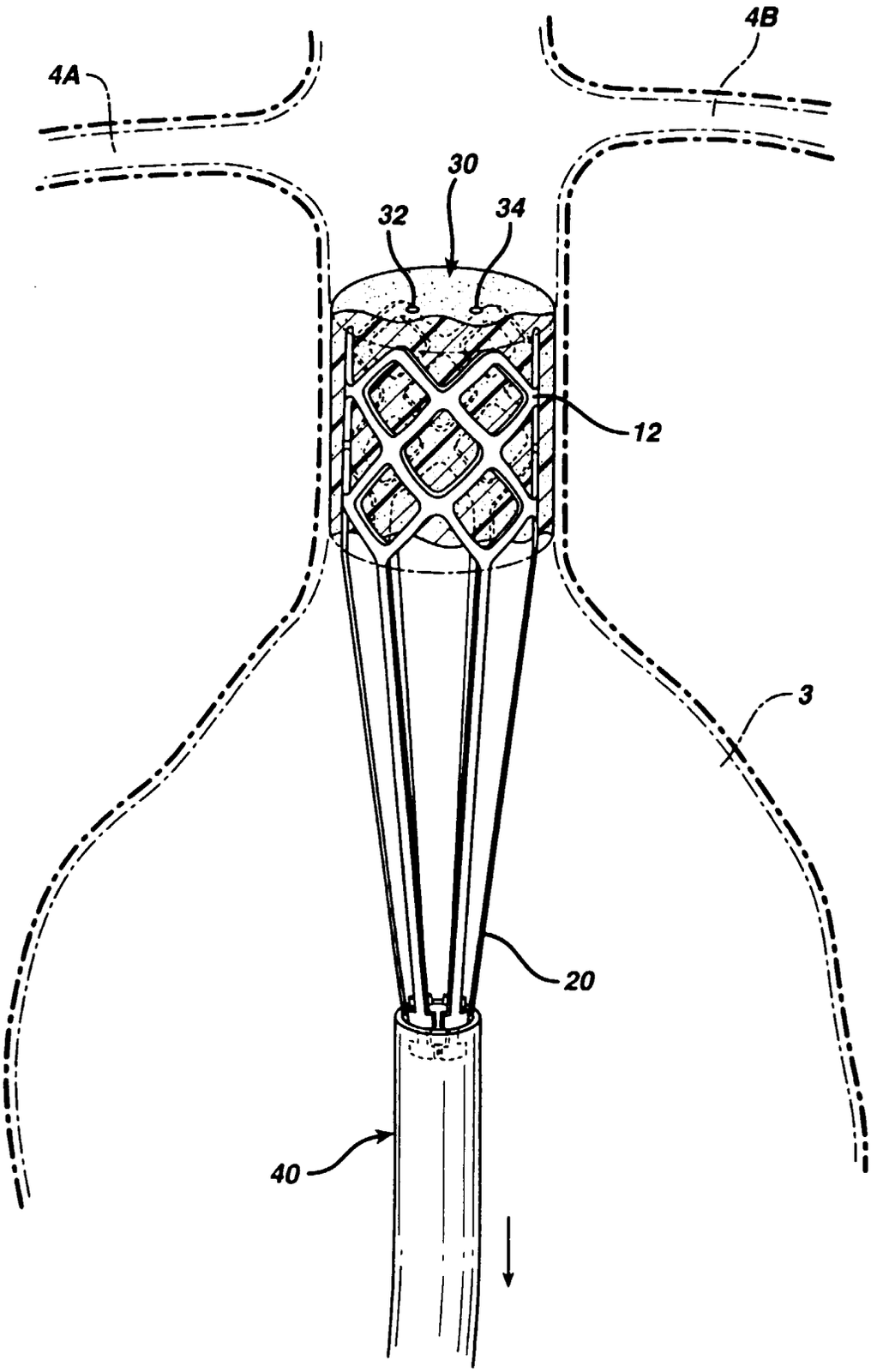


FIG. 5

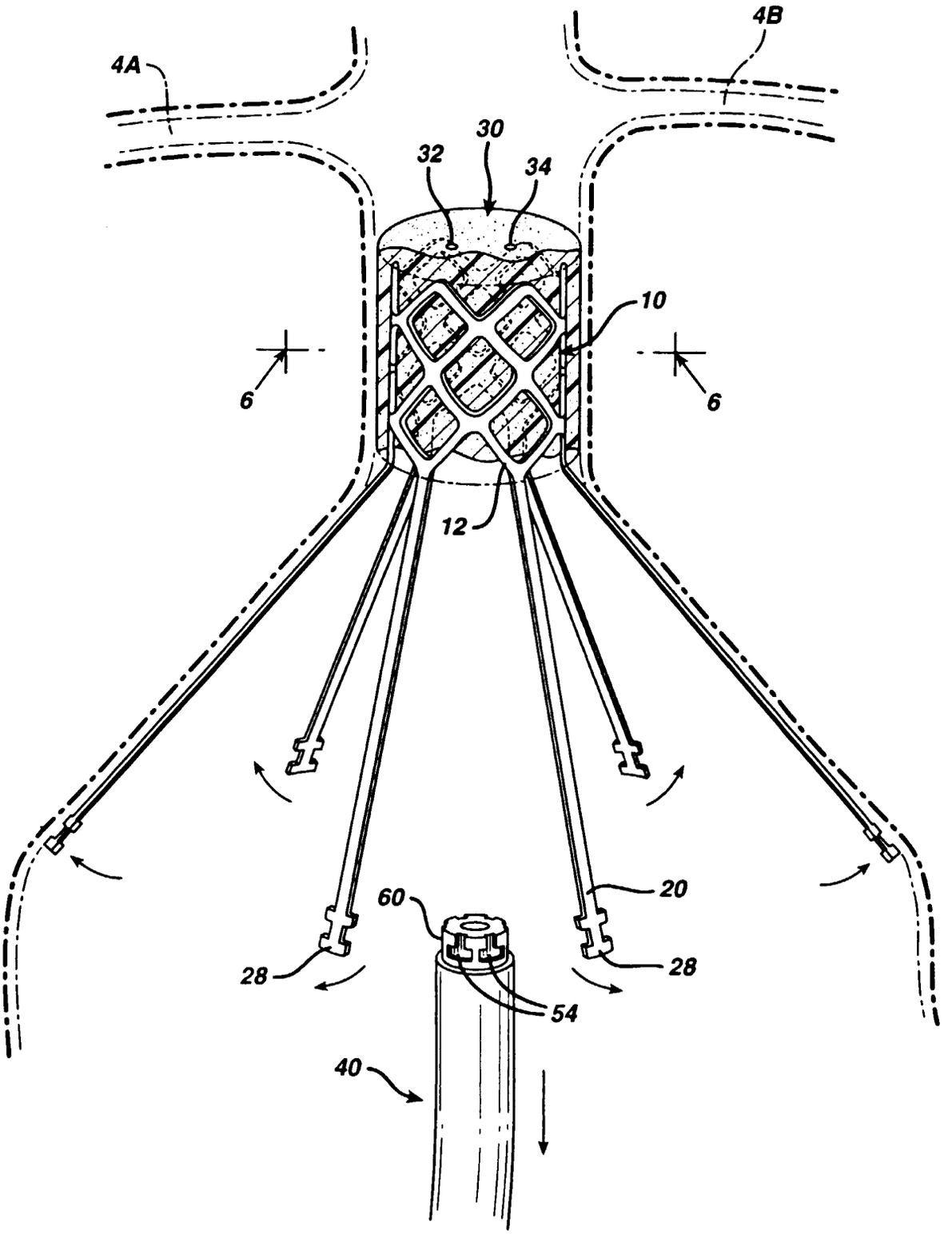


FIG. 6

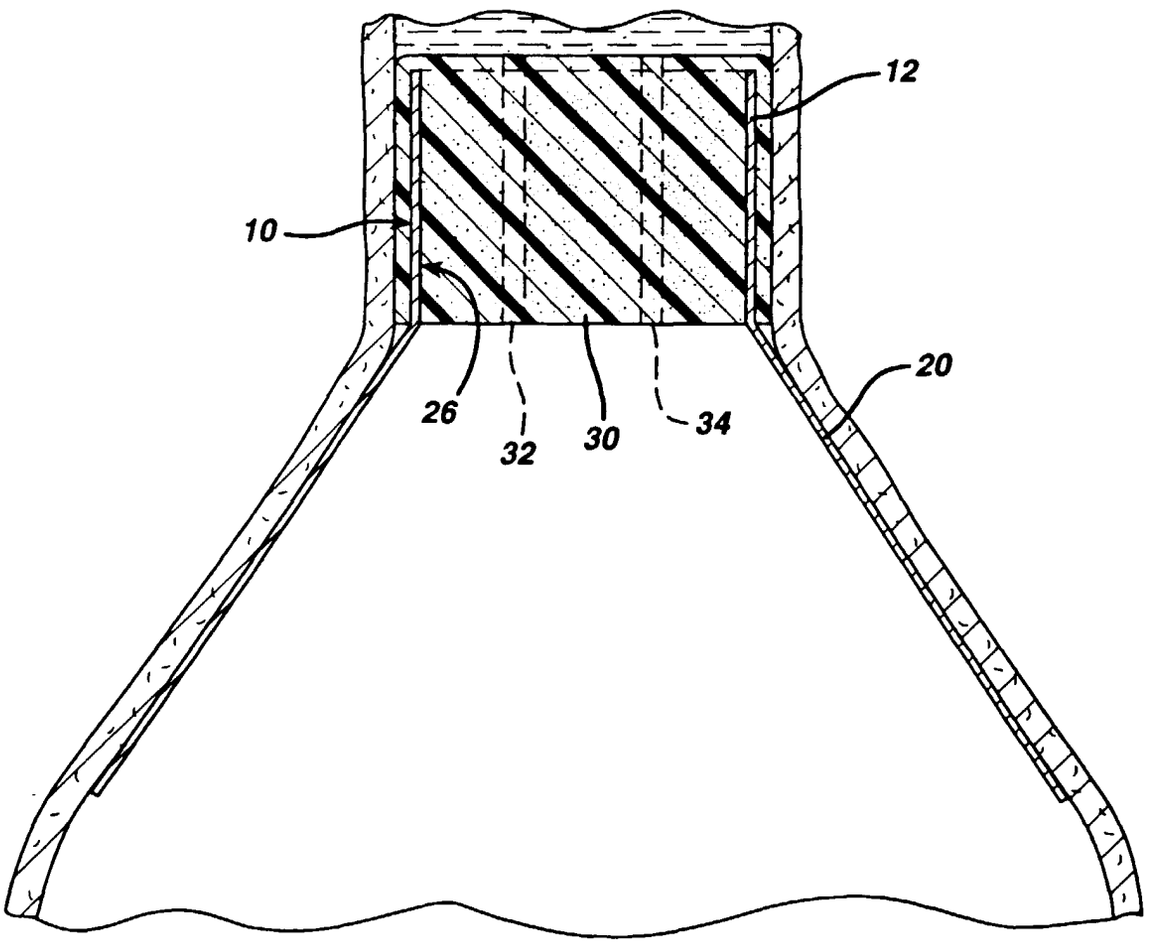


FIG. 7

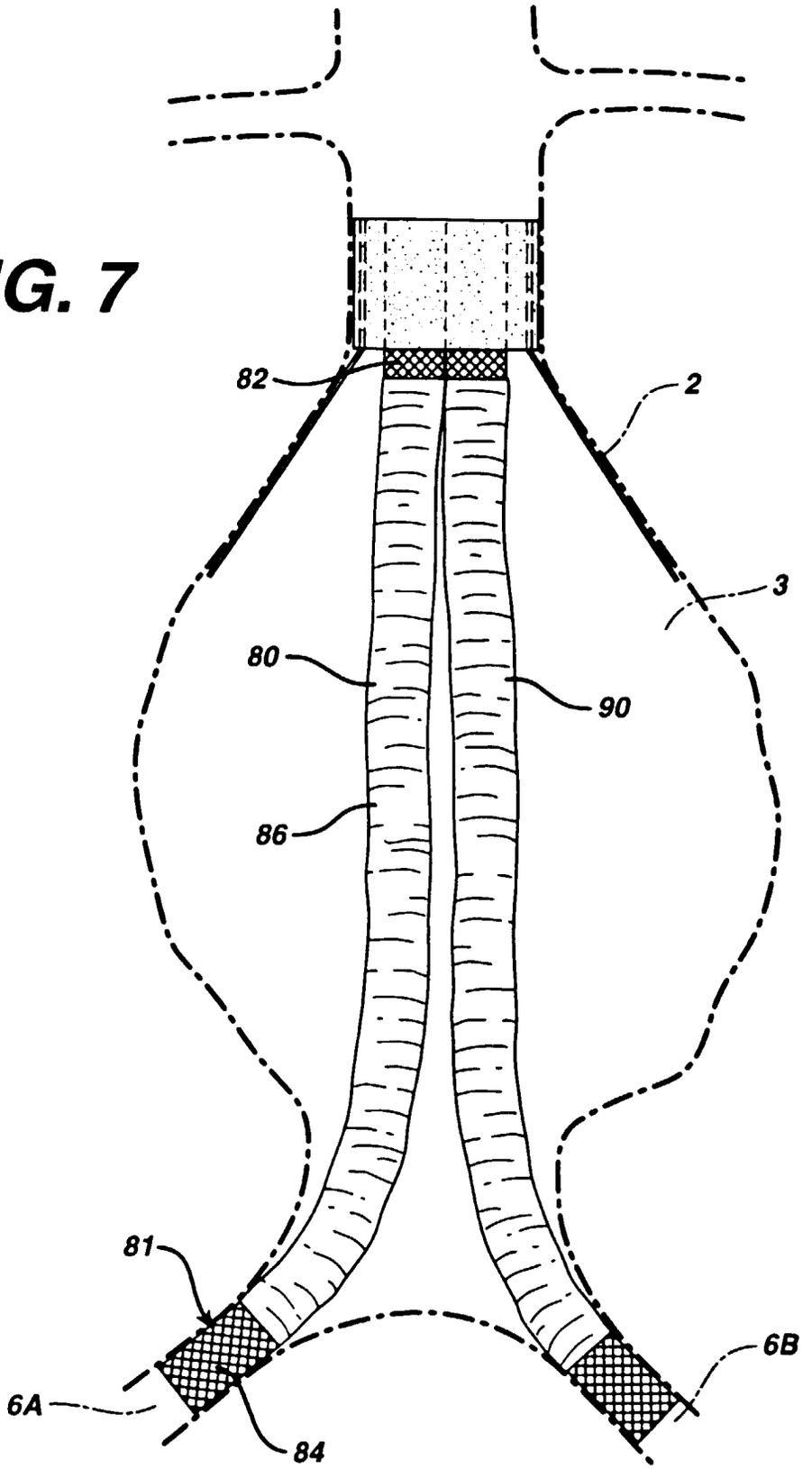
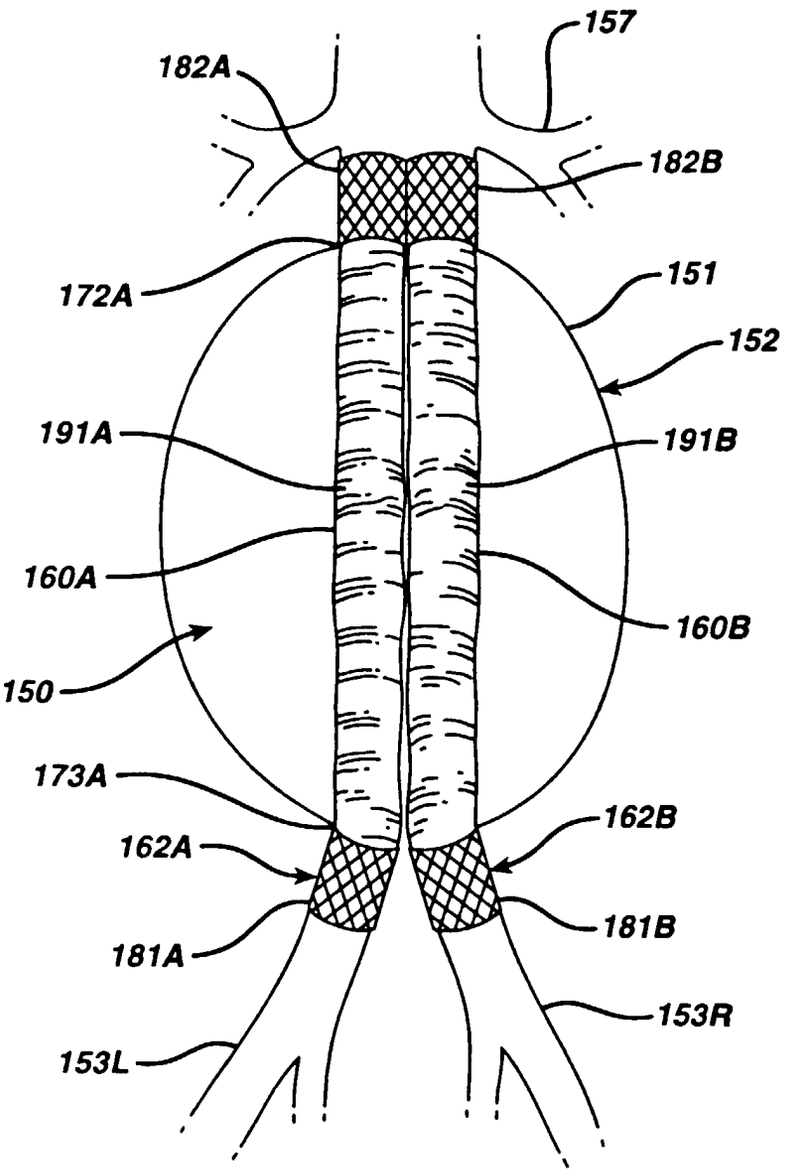


FIG. 8 PRIOR ART



European Patent Office

EUROPEAN SEARCH REPORT

Application Number
EP 99 30 8865

DOCUMENTS CONSIDERED TO BE RELEVANT

Category	Citation of document with indication, where appropriate, of relevant passages	Relevant to claim	CLASSIFICATION OF THE APPLICATION (Int.Cl.7)
A	EP 0 579 523 A (CELSA LG) 19 January 1994 (1994-01-19) * abstract; figure 17 *	1	A61F2/06
A	WO 98 23322 A (BOSTON SCIENTIFIC CORPORATION) 4 June 1998 (1998-06-04) * page 13, line 12 - line 18; figure 2 *	1	

TECHNICAL FIELDS SEARCHED (Int.Cl.7)

A61F

The present search report has been drawn up for all claims

Place of search	Date of completion of the search	Examiner
THE HAGUE	9 February 2000	Smith, C

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09-02-2000

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